

Lab 02: Image Processing Fundamentals

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Technical Understanding

One of the most interesting technical takeaways from this lab was learning that computers represent digital images as a multidimensional matrix of numbers. A color image is essentially a 3D matrix with height, width, and color channels. Each value can range from 0 to 255, where each number represents a color. It was interesting to learn how basic math operations can translate into visual effects. For example, adding a constant value to an entire array to increase the brightness. The most challenging techniques to understand were convolution and neighborhood operations. Point operations can process one pixel independently, but visualizing how a 3x3 kernel slides across an image matrix to average or compute requires a deeper understanding of the underlying math.

Connections & Applications

The lab activities showed how classic computer vision tools serve as a bridge to modern Artificial Intelligence (AI) tools covered in class, such as Nano Banana. The lab demonstrated how AI image processing connects to the fundamental operations of computer vision. Edge detection and texture analysis utilizing OpenCV filters mirror the feature extraction layers in Convolutional Neural Networks (CNNs). The biggest difference is that AI tools automatically learn millions of optimal filters from available datasets, whereas manually hand-coded matrices are limited by the user's input. Histogram equalization can be helpful for medical imaging to enhance X-ray contrast. In a future project, I could combine approaches from traditional computer vision to clean and preprocess an image before feeding it into a neural network for object recognition, which could help improve accuracy.

Personal Reflection

The topic that interests me most is the connection between traditional computer vision and modern AI. Learning how fundamental math operations serve as the building blocks for CNNs was fascinating. Traditional computer vision requires manually designing and applying matrices to extract features, whereas AI automatically learns millions of optimal filters from data. I am curious to learn how computer vision algorithms can be used to preprocess and clean data before it is fed into an AI model.